Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023
Comprehensive Exam (EC-3 Regular)

Course No. : AIMLCZG512 **Course Title** : Deep Reinforcement Learning
Nature of Exam : Open Book **Weightage** : 30% **No. of Pages** = 3 ; **No. Of Questions =** 4;
Duration : 150 mins; **Date of Exam** :

Note to Students:

1. Answer all the questions. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
2. Write answers neatly in A4 papers, scan and upload.
3. Write your name and sign at the end of all the pages.
4. Assumptions made if any, should be stated clearly at the beginning of your answer.
5. Refer to the Honor Code published at the course page.

Question -1 **[10 Marks]**

A reinforcement learning agent helps a healthcare professional choose between two treatments for patient recovery. At each visit, the agent selects one of the two actions: Medication(M), Therapy (T) where medication refers to standard drug-based treatment and therapy refers to physical rehabilitation approach. The agent assumes a Multi-Armed Bandit model where feedback (reward) is either **+1** (Successful recovery) or **-1** (No improvement). The corresponding dataset comprising 8-time steps is given below:

Time Step (t)	**Action Taken**	**Reward (R)**
1	M	+1
2	M	-1
3	T	+1
4	T	+1
5	T	-1
6	M	+1
7	T	-1
8	M	*r*

a) Using the UCB formula with c = 1.0, what is the maximum reward value r for which the UCB score of "Therapy" exceeds the UCB score of "Medication" at the 9th decision point? **(4 Marks)**

b) Suppose the agent uses a non-stationary approach for computing Q instead of sample averages, with α =0.5. Using Q_1 (M) =0, compute Q(M) at t =9 as a function of r. **(4 Marks)**

c) Comment on the Q(M) comparing that of part (a). How sensitive is the agent's treatment choice to r? **(2 Marks)**

Question -2 **[10 Marks]**

Scenario: A mobile robot operates in a warehouse with three battery states: Low (L), Medium (M), and High (H). At each decision point, the robot can either Continue Working (C) or Go Recharge (R).

MDP Specification:

States: {Low (L), Medium (M), High (H)} ; **Actions:** {Continue, Recharge}; **Discount factor:** $\gamma = 0.9$; **Current value estimates:** V(L) = 1, V(M) = 1, V(H) = 1.

State transitions :

Current State	Action	Next State	Probability	Reward
High (H)	Continue	High (H)	0.7	+2
High (H)	Continue	Medium (M)	0.3	+2
High (H)	Recharge	High (H)	1.0	−1
Medium (M)	Continue	Medium (M)	0.6	+1
Medium (M)	Continue	Low (L)	0.4	+1
Medium (M)	Recharge	High (H)	1.0	−2
Low (L)	Continue	Low (L)	1.0	−5
Low (L)	Recharge	Medium (M)	1.0	−3

a) **[2 Marks]** The robot currently follows policy π where π(Continue | M) = 1. Write the Bellman expectation equation for V_π(M) and calculate its value using the given estimates.

b) **[4 Marks]** The robot starts with the following initial policy π_0: π_0(Continue | H) = 1.0 ; π_0(Continue | M) = 1.0 ; π_0(Recharge | L) = 1.0. Given: V^0(L) = -10, V^0(M) = 5, V^0(H) = 8. Determine the improved policy π_1 at state M by computing: Q_{π_0}(M, Continue) ; Q_{π_0}(M, Recharge) . Should the policy at state M change? What is the new policy π_1(a|M)?

c) **[1 Marks]** The robot executes the policy π (Continue at M, Recharge at L) and generates the following episode with $\gamma = 0.9$:

Episode: (M, Continue,+1, M) →(M, Continue, +1, L) →(L,Recharge, -3,M) → (M, Continue, +1, Terminal).

Using First-Visit Monte Carlo, calculate the estimated value V(M) from this single episode. Show the return calculation with proper discounting.

d) **[3 Marks]** The robot now uses an exploratory behaviour policy b where: b(Continue | M) = 0.7; b(Recharge | M) = 0.3 The target policy π is deterministic: π(Continue | M) = 1.0 Consider the following episode under behavior policy b with $\gamma = 0.9$:

Episode: (M,Continue,+1,M) → (M,Recharge,-2,H) → (H, Continue, +2, Terminal).
Calculate:

d-1) The importance sampling ratio ρ from the first occurrence of state M until termination
d-2) The return G from state M
d-3) Explain how this ratio would be used in weighted importance sampling to update V(M)

Question -3 **[1+4+2+1+2= 10 Marks]**

Scenario: A personalized AI tutor platform curates tasks of varied difficulty level for practice as {Medium (M), High (H)} based on learner mastery categorized into {Weak (W), Partial (P), Strong (S)} and motivation level observed from their usage statistics including {Engaging (E), Disengaging (D)}.
Rewards:
For the Observation At time step t0 : (S, A, S`, R) = (S1: (P, D), H , S2 = (P, D) , +2) , answer the following questions sequentially.

a) Transform the states (P, D), (P, E) using the following coarse coding:
Features : (F1, F2, F3) , $\phi(S) = [F1\ \ F2\ \ F3]^T$

$$F1 = \begin{cases} 1, & \textit{Mastery level is Partial or Strong} \\ 0, & \textit{otherwise} \end{cases} \qquad F2 = \begin{cases} 1, & \textit{Motivation level is Disengaging} \\ 0, & \textit{otherwise} \end{cases}$$

$$F3 = \begin{cases} 1, & \textit{Mastery level is Partial and Motivation level is Disengaging} \\ 0, & \textit{otherwise} \end{cases}$$

b) Use the function approximator $\pi(\_|S) = \theta^T\phi(S)$ to compute the probability of choosing the difficulty level of tasks. Assume $\theta^T = [0.2 , 0.1 , 0.05]$ for High-difficulty task (H) and $\theta^T = [0,\ 0,\ 0]$ for Medium-difficulty task (M).
Using the given model, perform a one step policy gradient update with α=0.05 for the given observation to reinforce the model parameters with fixed baseline value of 1.

c) Compare the agent's behavior when α=0.05 vs α=0.7. Comment on its effect on the speed and stability of the learner.

d) In above subpart b) what will happen if the baseline is not considered? Explain w.r.t its impact on given scenario ie., AI tutor only.

e) Assume the Actor Critic algorithm (eg.,A2C) is chosen to be applied to the given use case. Briefly state the role of actor vs critic w.r.t the given scenario ie., AI tutor only.

Question -4 **[10 Marks]**

(a) During the Selection phase of Monte Carlo Tree Search, a node corresponding to state s has two available actions, a_1 and a_2. The stored statistics are:

Action	Visits $N(s,a)$	Mean value $Q(s,a)$
a_1	4	0.60
a_2	16	0.65

Compute the UCB score for both actions. Which action will be selected next? Justify numerically **[3 Marks]**

(b) **[2 Marks]** The company trains a value network for leaf node evaluation using Deep Q-Network (DQN**)** principles. State two key techniques DQN uses to stabilize deep neural network training and briefly explain why each addresses instability.

(c) **[2 Marks]** The MCTS algorithm is inspired by AlphaGo and AlphaZero**.** State one key difference between AlphaGo and AlphaZero regarding their training data. Why does this make AlphaZero more generalizable to other domains?

(d) The company uses imitation learning with expert demonstrations of gameplay.

(d-1) [1.5 Marks] Write the loss function for behavior cloning when training policy $\pi_\theta(a|s)$. State one major failure mode when the policy encounters unseen states.

(d-2) [1.5 Marks] How does DAGGER modify the data collection process compared to behavior cloning? Why does this help the policy handle its own mistakes?